MILITARY SPECIFICATION

RACK, BOMB, MULTIPLE PRACTICE A/A 37B-3

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification contains the requirements for the procurement of a multiple, practice bomb rack, designated type A/A 37B-3. The bomb rack consists of a body assembly having the designation MAK-53/A37B-3, and six release assemblies having the designation SUK-18/37B-3.

1.2 Classification - The equipment covered by this specification shall be of one type designated as Rack, Bomb, Multiple Practice A/A 37B-3 herein after referred to as the rack.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATION

Military

MIL-P-116	Preservation, Methods of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-T-7743	Testing, Store Suspension Equipment, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

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MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS3107	Connector, Plug, Electric, Quick Disconnect
MS35058	Switch, Toggle, Single Pole, 1-Hole Mounting

AIR FORCE - NAVY AERONAUTICAL BULLETINS
ANA 445 - Engineering changes to Weapons, Systems, Equipments and Facilities.

DRAWINGS

Bureau of Naval Weapons

DL 64A81-A1	Drawing List A/A 37B-3 Multiple Bomb Rack

2.2 Availability of documents

(a) When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCI, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

(b) Copies of applicable drawings may be obtained upon application to the Central Technical Documents Office, Naval Ordnance Plant, Louisville, Kentucky.

3. REQUIREMENTS

3.1 Preproduction - The rack furnished under this specification shall be a product which has been tested and has passed the preproduction tests specified herein.

3.2 Acceptance - Each rack shall be subject to and shall pass the acceptance tests specified herein.

3.3 Material - Material shall be in accordance with that shown on drawings listed on DL 64A81-A1.

3.4 Construction - The rack shall be constructed in accordance with the drawings listed on DL 64A81-A1. During construction of the rack, the following parts shall be precisely located to permit satisfactory assembly.

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Part Name	Part Number
Connector	97-5105-14S-55
Cable	64A81B44
Plug Assembly	64A81B44
Bracket, Solenoid Support	64A81C42
Block Support	64A81C16
Bulkhead Assembly	64A81D17

All adjustments shall be as called out on applicable drawings.

3.5 Performance - The rack, when mounted on an aircraft for which designed, shall be capable of suspending and releasing stores in the 0-99 pound class.

3.6 Input Power - The input power to operate and test the rack shall be from a 28VDC power source with characteristics and limits as defined in Standard MIL-STD-704.

3.7 Operational environment - This equipment shall perform satisfactorily under all applicable conditions specified in Specification MIL-T-7743.

3.8 Identification of product -

3.8.1 Nameplates - Nameplates shall be in accordance with applicable drawings which are listed on DL 64A81-A1.

3.8.2 Equipment, assemblies, and parts - Equipment, assemblies and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.9 Workmanship - The bomb rack shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from defects, burrs, and sharp edges; accuracy of dimensions and marking of parts and assemblies; the alignment of parts and tightness of assembly screws and bolts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contractor purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assume supplies and services conform to prescribed requirements.

4.1.1 Test Samples - Racks used for preproduction and sampling tests shall not be considered as equipments under the contract. They are to be supplied as equipments over and above the quantity specified for delivery. These racks will not be returned to the manufacturer.

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4.2 Classification of Tests

(a) Preproduction tests - Preproduction tests are those tests conducted on racks representative of the equipments to be supplied under the contract for the purpose of determining that the sample items meet all the requirements of the applicable specifications.

(b) Acceptance tests - Acceptance tests are those tests accomplished on racks submitted for acceptance under the contract and shall consist of the following:

1. Individual tests - Individual tests are those tests conducted on each rack submitted for acceptance under contract.

2. Sampling tests - Sampling tests are those tests accomplished on a quantity of racks selected as specified herein and which have previously passed individual tests.

4.3 Preproduction tests -

4.3.1 Preproduction test sample - Two complete racks shall be delivered, at the contractors expense, to a specified government laboratory for preproduction tests.

4.3.2 Tests - The preproduction tests shall consist of all the tests of this specification as described under section on test methods. The tests shall be conducted on each of two racks according to the schedule shown on Table I.

TABLE I
PREPRODUCTION TESTS

TEST	SAMPLE NUMBER 1	2	APPLICABLE TEST PARAGRAPH
Assemblies and Parts Test	x	x	4.6.1
Examination of Product	x	x	4.6.2
Magnetic Particle Inspection	x	x	4.6.3
Release Assembly Fit	x	x	4.6.4
Latch Release Mechanism Force	x	x	4.6.5
Release Sequence Test	x	x	4.6.6
Physical Characteristics	x	x	4.6.7

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TABLE I (Cont'd)

TEST	SAMPLE NUMBER		APPLICABLE TEST PARAGRAPH
	1	2	
Electrical Characteristics	x	x	4.6.7
Lag or Time of Release Determination	x		4.6.7
Performance Under Icing Conditions	x		4.6.7
Shock Test		x	4.6.7
High G Release Test	x		4.6.7
Sand Test	x		4.6.7
Static Tests	x	x	4.6.7
Salt Spray Test		x	4.6.7
Life Test	x		4.6.7
Random Vibration		x	4.6.8

4.4 Acceptance tests - The acceptance tests shall consist of individual tests and sampling tests.

4.4.1 Individual tests - Each rack submitted for acceptance under contract shall be subject to the following tests as described under section on test methods.

TEST	APPLICABLE TEST PARAGRAPH
Assemblies and Parts Test	4.6.1
Examination of Product	4.6.2
Magnetic Particle Inspection	4.6.3
Release Assembly Fit	4.6.4
Latch Release Mechanism Force	4.6.5
Release Sequence Test	4.6.6
Physical Characteristics	4.6.7
Electrical Characteristics	4.6.7

4.4.2 Sampling tests -

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4.4.2.1 Sampling Plan - Sample racks shall be delivered, at the contractors expense, to a specified government laboratory for testing. Samples shall be subjected to all the tests of 4.6.7 and shall be selected at random by the local authorized government inspector or representative in accordance with the following:

Quantity Offered For Acceptance	Quantity to be Selected For Sampling Tests
First 10	1
Next 50	1
Next 100	1
Next 200	1
	1 for each additional 200 or fraction thereof

A lot shall consist of racks produced under essentially the same manufacturing conditions, and submitted for inspection at the same time. The unit of inspection shall be one rack.

4.5 Test conditions - Unless otherwise specified herein, all tests shall be conducted at the conditions specified in Specification MIL-T-7743.

4.6 Test Methods -

4.6.1 Assemblies and parts tests - Tests on assemblies and parts shall be performed as shown on applicable drawings listed on DL-6481-A1.

4.6.2 Examination of product.- Each rack shall be inspected to determine compliance with the requirements of this specification not covered by test methods.

4.6.3 Magnetic particle inspection - All steel parts shall be subjected to the magnetic particle inspection in accordance with Specification MIL-I-6868. There shall be no evidence of cracks or other defects which might affect subsequent use.

4.6.4 Release assembly fit - Assembly of the release assemblies, part number 64A81D5 to the body assembly, part numbers 64A81H4, 64A81C10, and 64A81B25 shall be determined by:

(a) Interchangeability of release assemblies and body assemblies and a force free fit.

(b) A positive electrical connection

(c) Alignment of the longitudinal centerline of each release assembly parallel with the longitudinal centerline of the body assembly within one-half degree.

4.6.5 Latch release mechanism force - The allowable force required to latch the release mechanism shall not exceed 39 pounds.

4.6.6 Release sequence test - The release sequence test shall be performed as follows:

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(a) Use a suitable test cable with a receptacle, Standard MS3107A-16S-1S, and a monitor switch similar to Standard MS35058-30 (See Figure 1). Use 28 VDC which is in accordance with Standard MIL-STD-704. Apply plus 28 VDC to pin B, with pin E grounded in the plug, Standard MS3102A-16S-1P, on the body assembly. Insure that the receptacle fully seats on the plug, and that the retention bails lock the receptacle. Position the monitor switch in the "OFF" position.

(b) Latch each of the release mechanisms.

(c) Set the selector switch to the "NUMBER 1" position.

(d) Actuate the monitor switch six times. The release mechanism shall actuate the free and positive action and in the sequence shown on the nameplate. In addition, the selector shall advance one position with each actuation.

4.6.7 Tests of Specification MIL-T-7743 - The following tests of Specification MIL-T-7743 shall be performed on a complete assembled rack with procedures as outlined therein:

Non-Destructive Tests

Physical Characteristics
Electrical Characteristics
Lag or Time of Release Determination
Performance under Icing Conditions
Shock Test
High G Release Test
Sand Test
Static Test

Destructive Test

Salt Spray Test

4.6.8 Random Vibration - Random Vibration shall be applied to each of the three mutually perpendicular axes of the bomb rack. When using manual equalization, the random vibration shall be applied at 20% of test values for sufficient time to adjust the filters. The equalized frequency spectrum shall be continuous and uniform within $\pm$ 3db over the frequency range of 20 to 2000 cps as determined by analysis using a bandwidth of 25 cps or less (and a sufficient sweep speed and appropriate averaging for consistent and accurate analysis, if applicable). Random vibration at .05 g/cps over the 20 to 2000 cps bandwidth (10g rms overall) having Gaussian distribution and a peak to rms G ratio of 3 to 1 shall be applied to the unloaded bomb rack for a period of 40 minutes in each axis. The rack shall be unlatched and released for unloaded vibration. There shall be do damage and the rack shall not fail to function as a result of this test.

4.6.8.1 Loaded Vibration - The random vibration test shall be repeated with the bomb rack loaded. The loaded vibration shall be conducted with the heaviest store suspended on each station and properly swaybraced. After 20 minutes of loaded vibration, the stores shall be released while the vibration is continued. Vibration shall then be

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stopped and the bomb rack shall be reloaded. The Vibration Test shall then be continued for another 20 minutes. Vibration shall then be stopped. The stores shall be released to insure that the vibration has not caused the item to jam. The equalized spectrum shall be monitored during the test and shall be readjusted as necessary. There shall be no damage and the bomb rack shall not fail to function as a result of this test.

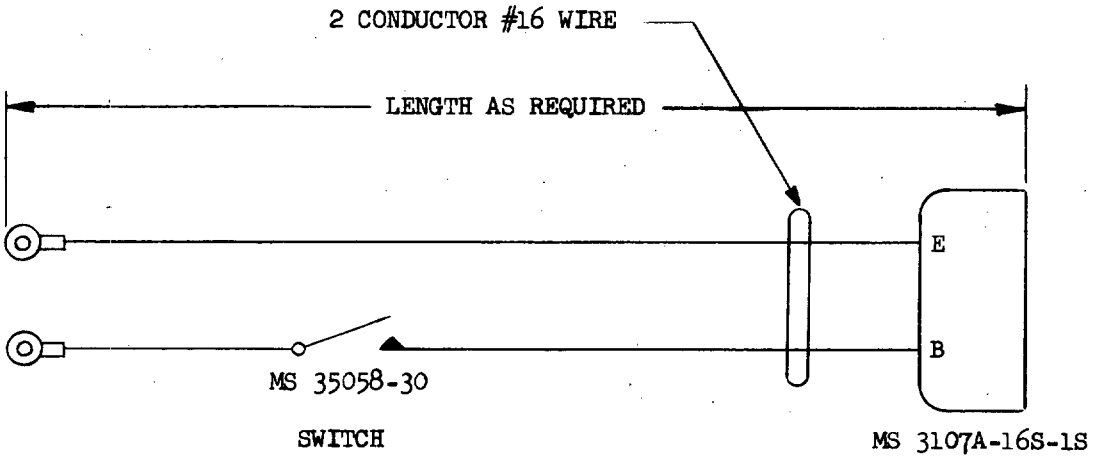


Figure 1 - Test Cable

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - The rack shall be unit packaged in accordance with Specification MIL-P-7936, levels A or C as specified (see paragraph 6.2), without the use of preservative compound.

5.1.1 Level A - For level A packaging, preservation shall be in accordance with Specification MIL-P-116, Method II without preservative compounds.

5.1.2 Level C - The racks shall be preserved and packaged individually in accordance with manufacturer's commercial practices.

5.2 Packing - Packing shall be in accordance with Specification MIL-P-7936, level A, B, or C as specified in contract or order.

5.3 Marking - Interior and exterior containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use - The rack covered by this specification is intended for practice bombs using the stores of the 0-99 pound class. The rack is designed to be used with applicable adapter kits for each type of aircraft as shown on DL 64A81-A1.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Number of racks required.

(c) Required level of packaging and packing (see para. 5.1).

(d) Place of conducting Preproduction Test (Bench and Flight Tests).

(e) Installation instructions and handbooks.

(f) Quantity and drawing numbers of Kits (if required)

6.3 Adherence to drawings and specifications - Conflict Eliminations - Insofar as the contractor is required by provisions of this specification to adhere rigidly to the drawings listed in paragraph 2.1 as applicable, which may make it impractical in some instances, also to comply with the performance and test requirements specified herein, then, in that event, the contractor should promptly notify the procuring activity of such fact, and appropriate changes may be made in the applicable drawings, parts of performance requirements as authorized by the procuring activity. Material and design changes shall be submitted to procuring activity using the procedure specified in Air Force - Navy - Aeronautical Bulletin ANA-445.

6.5 Precedence of documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.6 Provision for preproduction tests - Quantity production on a particular contract shall be withheld until the preproduction sample has been pronounced satisfactory by the procuring activity. When a contractor is in continuous production of these racks from contract to contract, submission of additional preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirements of production testing.

6.6.1 All racks identical to preproduction racks - It is understood that the racks supplied under contract will be identical in every respect to the corresponding approved preproduction sample and to the drawings except for changes which have previously been approved by the Government.

6.7 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents referenced in Section 2, unless specified in the contract or order.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-R-23818(WP) RACK, BOMB, MULTIPLE PRACTICE A/A 37B-3

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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